VIMD-Net: Physically Supervised Representation Allocation for AMC under Vehicular-Doppler TDL Simulations

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***Abstract*—Automatic modulation classification (AMC) under mobile multipath is impaired by structured interference that overlaps modulation cues in time and frequency. This paper develops VIMD-Net, a phase-preserving representation-allocation network whose environment-conditioned three-way mask allocates target-power-dominant, jammer-power-dominant, and unexplained-or-power-ambiguous evidence on the physical complex short-time Fourier transform lattice. A fixed training-only teacher is constructed from independently tracked target, jammer, and unexplained simulation components; dual task branches, auxiliary jammer/link supervision, and exact-source cross-condition contrastive learning constrain the route semantics. The evaluation is locked to source-disjoint 3GPP TR 38.901 TDL-profile simulations parameterized by vehicular Doppler and structured jammers, with independently held-out jammer, speed, and channel factors, single-view inference, and source-paired statistical comparisons. Unanchored in-taxonomy modulated cochannel collisions are excluded because their single-label target is not identifiable from a mixture alone. Quantitative claims are intentionally withheld in this internal-review build until the frozen multi-seed evidence gate is passed; no SDR, field, or deployment claim is made.**

***Index Terms*—Automatic modulation classification, vehicular communications, interference robustness, representation disentanglement, soft masks, contrastive learning.**

Evidence lock. This build is for internal scientific review. Cells marked "pending" are not zeroes and must not be interpreted as experimental outcomes. Smoke runs and fixed-batch overfit diagnostics are excluded from paper claims.

I. Introduction

AUTOMATIC modulation classification enables a receiver to infer the modulation of an unknown waveform without transmitter-side control information. It supports spectrum monitoring, blind receiver configuration, interference diagnosis, and adaptive signal processing. Deep AMC models have advanced from convolutional classifiers [1], [2] to complex-valued [3], dual-stream [4], causal-convolutional [5], and Transformer-based [6], [7] representations. Efficiency, hardware impairment, and channel robustness have also been studied [8]–[10].

A mobile receiver poses a different failure mode from the usual additive-white-Gaussian-noise benchmark. Mobility produces time-varying multipath and Doppler, while concurrent transmitters and unintended emitters create narrowband, impulsive, swept, partial-band, and broadband interference. The target and jammer can therefore share both time and frequency support. Direct classifiers must learn modulation and nuisance factors from a single polluted representation. A hard suppression stage is also unsafe: a cell with high jammer power may still contain target power.

Recent work narrows individual parts of this gap. Domain adaptation improves cross-domain AMC [11], [12]; masked and contrastive learning improves representation robustness [13]–[15]; interference-tolerant feature extraction directly addresses corrupted observations [16]; and overlapped-signal classification studies simultaneous emitters [17]. Consequently, neither "soft masking," "contrastive AMC," nor "multi-task AMC" is a defensible novelty claim by itself. What remains insufficiently tested is whether a mask can be assigned an auditable physical meaning, explicitly retain ambiguous target–jammer cells, and improve modulation invariance under independently held-out vehicular and jammer conditions.

This work addresses that question with VIMD-Net. It is derived from a broader interference-adaptation concept, but replaces template matching, binary component deletion, and rule-based label correction with differentiable, task-constrained representation allocation. The paper deliberately does not claim waveform recovery, strict source separation, protocol identification, modulation recommendation, or open-set recognition.

The contributions are threefold.

- We introduce an environment-conditioned three-way mask on the exact complex-STFT lattice. It allocates target-power-dominant, jammer-power-dominant, and unexplained-or-power-ambiguous evidence with a learned overlap gate and a bounded additive bypass.
- We construct a fixed, component-traceable simulation-component power teacher from target, jammer, and unexplained power on that same lattice. Jensen–Shannon mask supervision and separate modulation, jammer, and link-quality objectives make the branch semantics experimentally falsifiable.
- During training, we pair two independently impaired observations of the same immutable source sequence and apply class-aware exact-source contrastive learning; inference uses one view. The accompanying protocol isolates jammer-family, speed, and channel-profile shifts and predeclares accuracy, calibration, source-paired significance, diagnostic mechanism variables, and complexity.

II. Related Work and Claim Boundary

A. Deep AMC

CNN-based AMC established that learned I/Q features can outperform hand-designed pipelines under matched synthetic conditions [1], [2], [18]. Later work preserved complex structure [3], fused spatial and temporal streams [4], reduced

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complexity [8], and improved causal or Transformer modeling [5]–[7]. These models form necessary direct-classification comparators; architectural depth alone cannot establish interference disentanglement.

B. Robustness, Masking, and Contrastive Learning

Channel-robust hand-crafted statistics [10], explicit hardware impairment compensation [9], domain adaptation [11], [12], masked contrastive learning [13], denoising masked autoencoding [19], and cross-SNR contrastive learning [15] address different nuisance sources. MFENet explicitly targets interference tolerance [16]; malicious-interference recognition has also been studied for MIMO observations [20]. Our narrow distinction is a three-route component-power teacher on the student's physical lattice, with the other physical condition of the exact source as the contrastive positive. Its target-power-dominant, jammer-power-dominant, and unexplained-or-power-ambiguous names are task assignments, not identified sources. We claim neither the first mask nor the first contrastive AMC model.

C. From Patent Concept to Scientific Hypothesis

The source concept follows an environment adaptation $\rightarrow$ interference suppression $\rightarrow$ modulation decision chain. The scientific implementation makes four material changes. First, a continuous environment embedding replaces a discrete jammer template database. Second, soft three-way allocation replaces hard threshold deletion. Third, a discriminative classifier replaces nearest-prototype and rule-forced label revision. Fourth, the jammer state is an auxiliary output and never overrides the observed modulation class. This separation prevents a link-adaptation recommendation prior from being confused with signal identification.

III. System Model and Learning Problem

A. Received Signal

For a length-N complex baseband window, let

$$x[n] = \mathcal{G}_\eta\left(\begin{array}{l} \displaystyle\sum_{\ell=0}^{L_s-1} h_s[n,\ell] s_m[n-\ell] \\ \displaystyle + \sum_{q=1}^{Q} a_q \sum_{\ell=0}^{L_{j,q}-1} h_{j,q}[n,\ell] j_q[n-\ell-\tau_q] \\ + w[n] \end{array} \right), \quad (1)$$

where s_m is a source sequence with modulation label m, h_s and $h_{j,q}$ are independently seeded time-varying channels, j_q is a structured interferer, and $\mathcal{G}_\eta$ collects receiver effects shared by the complete observation. Equation (1) permits a general Q, but the present evidence protocol admits exactly one structured jammer family per active view; multi-jammer compositional mixtures are outside scope. The generator retains the post-channel components

$$x = s + j + u, \qquad u = w + a_{\mathrm{rx}}, \quad (2)$$

after a common gain convention, where a_{rx} is the tracked receiver artifact. Realized powers, rather than requested scale factors, define

$$\mathrm{SNR} = 10\log_{10}\frac{\|s\|_2^2}{\|w\|_2^2}, \quad \mathrm{SIR} = 10\log_{10}\frac{\|s\|_2^2}{\|j\|_2^2}. \quad (3)$$

No-jammer samples carry a validity mask instead of a finite fictitious SIR.

The main task estimates $p_\theta(m \mid x)$. Training-only auxiliary targets are a multi-hot jammer vector and normalized SNR, SIR, and maximum Doppler. At inference, only x is required.

B. Exact-Source Paired Views

An immutable source identity contains the modulation, symbol sequence, and pulse-shaping realization. For source i, two views are

$$x_i^{(v)} = \mathcal{A}\Big(s_i;\, h_{s,i}^{(v)}, h_{j,i}^{(v)}, j_i^{(v)}, \eta_i^{(v)}\Big), \quad v \in \{1,2\}, \quad (4)$$

with independent channel, jammer, receiver, and noise seeds. All descendants of one source identity remain in one split. Thus the pair supplies a valid training positive without allowing bit-sequence leakage across train, validation, or test sets. Only one received view is supplied for each inference decision; the second view is used during paired training and source-aligned statistical comparison, not as test-time side information.

IV. VIMD-Net

A. Physically Aligned Spectral Front End

The input is represented as $X = [I; Q] \in \mathbb{R}^{2\times N}$. A deterministic, non-centered, full-spectrum complex STFT produces

$$Z = \mathrm{STFT}(I + \mathrm{j}Q) \in \mathbb{C}^{F\times T}. \quad (5)$$

The student context encoder receives

$$H = C_\phi(\mathrm{stack}\{\Re Z, \Im Z, \log(1+|Z|)\}), \quad (6)$$

where C_ϕ is a shallow 2-D convolutional residual encoder that preserves the $F\times T$ mask lattice. This is a phase-aware real implementation, not a claim of an end-to-end complex-valued neural network.

B. Environment-Conditioned Tri-Mask

Mean, standard deviation, and maximum pooling form

$$\mathrm{Stats}(H) = [\mu(H), \sigma(H), \max(H)], \quad (7)$$
$$e_c = E_\psi(\mathrm{Stats}(H)). \quad (8)$$

Feature-wise conditioning is

$$\widetilde{H} = \big(1 + \tfrac{1}{2}\tanh\gamma(e_c)\big) \odot H + \beta(e_c). \quad (9)$$

The model predicts

$$\tau_c = \tau_{\min} + (\tau_{\max} - \tau_{\min})\sigma(g_\tau(e_c)), \quad (10)$$

$$[M_s, M_j, M_o] = \mathrm{softmax}_{\mathrm{route}}\Big(G_\omega(\widetilde{H})/\tau_c\Big), \quad (11)$$

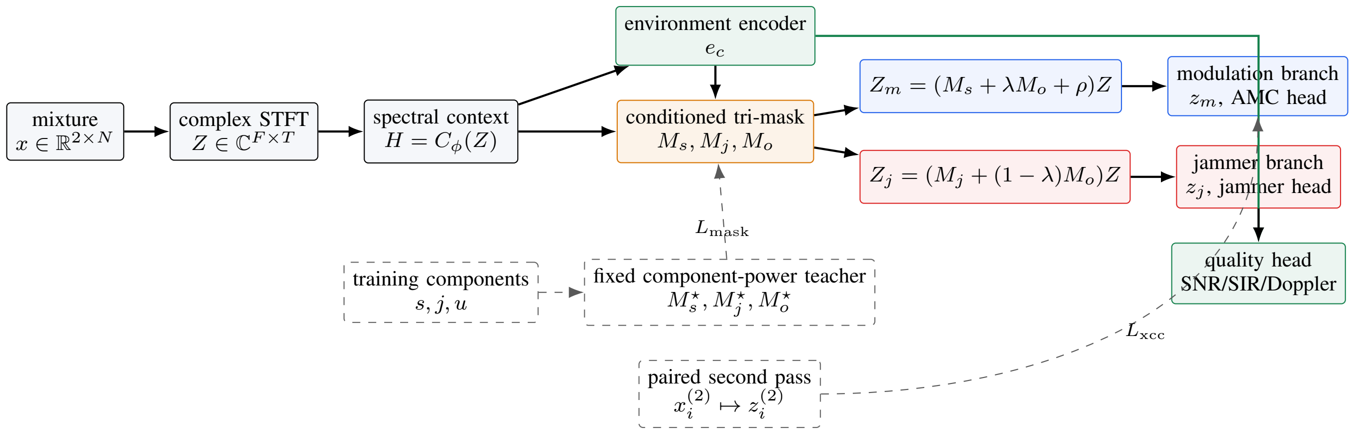

Fig. 1. Implemented VIMD-Net data flow. Solid paths are used at inference; dashed paths exist only during training. The masks act on the complex STFT itself, so the teacher and student share the same frequency–frame cells. The bounded additive ρ bypass intentionally makes the two branch weights non-conservative; whether it improves target-task robustness is tested by the predeclared A5–A7 comparison.

so the three nonnegative masks sum to one at every time-frequency cell. Here M_s, M_j, and M_o denote target-power-dominant, jammer-power-dominant, and unexplained-or-power-ambiguous allocations, respectively. These are component-power task routes, not identified sources.

Two bounded gates allocate the ambiguous route:

$$\lambda = \sigma(g_\lambda(e_c)), \tag{12}$$

$$\rho = \rho_{\min} + (\rho_{\max} - \rho_{\min})\sigma(g_\rho(e_c)), \tag{13}$$

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \tag{14}$$

$$Z_j = (M_j + (1-\lambda)M_o) \odot Z. \tag{15}$$

Importantly, λ, ρ, and τ_c are condition-dependent but are not constrained to be monotonic in SIR. Any such relationship must be measured rather than asserted.

C. Fixed Simulation-Component Three-Route Teacher

Let $P_s = |\mathrm{STFT}(s)|^2$, $P_j = |\mathrm{STFT}(j)|^2$, and $P_u = |\mathrm{STFT}(u)|^2$ on the same lattice as Z. Define

$$q_k = \frac{P_k}{P_\Sigma}, \quad P_\Sigma = P_s + P_j + P_u > 0, \quad k \in \{s, j, u\}, \tag{16}$$

For $v = q_s + q_j > 0$, let $r_s = q_s/v$, $r_j = q_j/v$, and $d = r_s - r_j$; when $v = 0$, set $r_s = r_j = d = 0$. The teacher is

$$M_s^\star = v[d]_+, \tag{17}$$

$$M_j^\star = v[-d]_+, \tag{18}$$

$$M_o^\star = q_u + v(1 - |d|). \tag{19}$$

Thus equal target–jammer power is assigned to uncertainty, while receiver artifacts and noise are always unexplained. The teacher is deterministic, has no trainable or exponentially averaged parameters, and is removed at inference. A zero-power cell is assigned $(0, 0, 1)$. The implementation uses the per-example guard $\epsilon_p = \max(10^{-8}\,\mathrm{mean}_{f,t} P_\Sigma, \mathrm{tiny})$: cells with $P_\Sigma \le \epsilon_p$ are assigned $(0, 0, 1)$, all other denominators are lower-bounded for numerical safety, and every cell is finally renormalized to the simplex. Thus the proposition below describes the ideal nonempty definition; the finite-precision implementation deliberately maps negligible-energy cells to uncertainty.

Proposition 1 (Simplex and common-scale invariance)**.** *For nonnegative component powers,* (17)–(19) *are nonnegative and sum to one wherever* $P_\Sigma > 0$*; the stipulated zero-power assignment is also on the simplex. In exact arithmetic, multiplying all three complex components by a common nonzero scalar leaves the masks unchanged.*

Proof. Since $r_s + r_j = 1$, $|d| \le 1$. Hence all three terms are nonnegative. Furthermore,

$$\begin{aligned} M_s^\star + M_j^\star + M_o^\star &= v|d| + q_u + v(1 - |d|) \\ &= v + q_u = q_s + q_j + q_u = 1. \end{aligned}$$

A common complex scale multiplies every P_k by the same positive constant, which cancels in q_k and all downstream quantities. □

Lemma 1 (Phase preservation)**.** *For a real nonnegative mask* $W[f,t]$*, the masked coefficient* $\widetilde{Z}[f,t] = W[f,t]Z[f,t]$ *has* $\angle\widetilde{Z}[f,t] = \angle Z[f,t]$ *whenever* $W[f,t] > 0$*.*

Proof. A positive real scalar changes only the magnitude of a complex number. At $W = 0$ the coefficient is suppressed and phase is undefined. □

The lemma motivates the phrase "phase-preserving allocation." It does not imply perfect reconstruction or source identifiability.

D. Dual Branches and Exact-Source Contrast

Independent spectral encoders yield $z_m = P_m(Z_m)$ and $z_j = P_j(Z_j)$. A linear head predicts modulation from z_m; a multi-label head predicts jammer presence from z_j; and a quality head predicts normalized SNR, valid SIR, and Doppler from $[e_c; z_j]$.

For a batch of paired views, let $\bar{z}_i^{(1)}$ and $\bar{z}_i^{(2)}$ be unit-normalized modulation embeddings. The valid comparison set for anchor i is

$$\mathcal{A}_i = \{i\} \cup \{k : y_k \neq y_i\}. \tag{20}$$

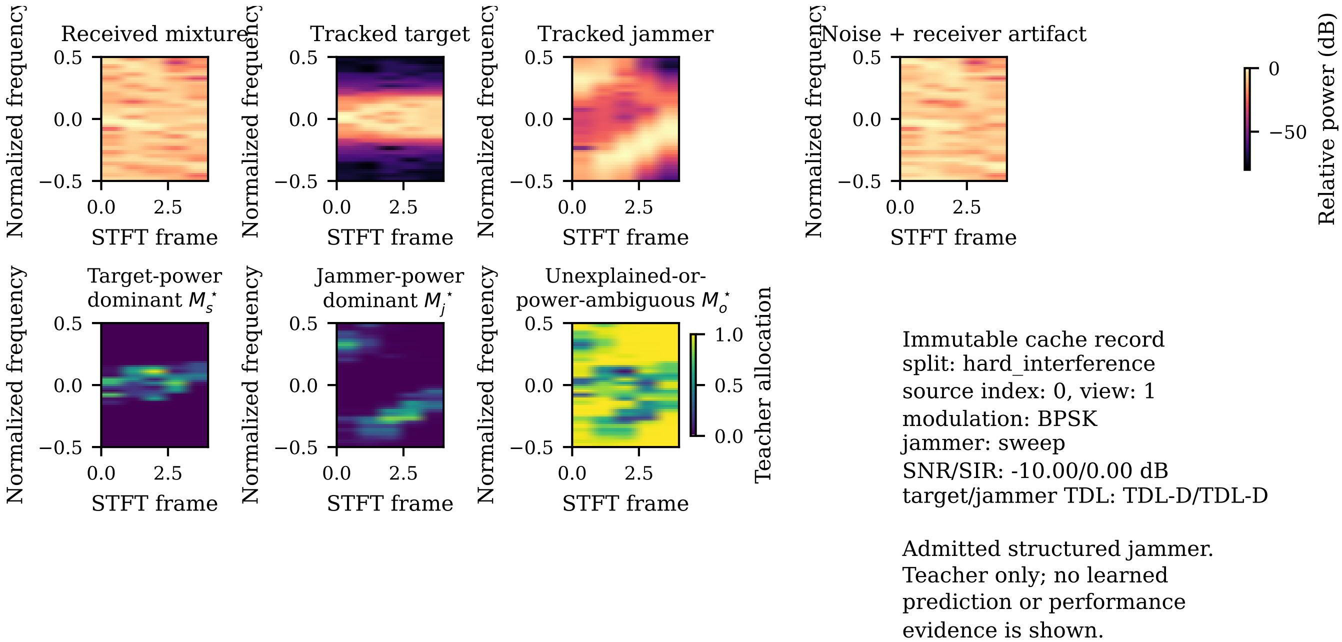

Fig. 2. Auditable fixed-teacher example from one immutable hard-interference cache record containing one admitted structured sweep jammer. The upper row shows the received mixture and independently tracked post-channel simulation components; the lower row shows the three component-power allocations on the identical STFT lattice. This deterministic illustration is neither a learned prediction nor performance evidence.

The diagonal is the other view of the exact same source; other sources of the same modulation are ignored. One direction of the loss is

$$\ell_i^{1\to 2} = -\log \frac{\exp(\bar{z}_i^{(1)\top}\bar{z}_i^{(2)}/\tau_x)}{\sum_{k\in\mathcal{A}_i}\exp(\bar{z}_i^{(1)\top}\bar{z}_k^{(2)}/\tau_x)}. \tag{21}$$

The cross-condition loss averages both directions. Unlike generic supervised contrastive learning [21], it neither treats all same-class signals as positives nor turns a different source of the same class into a false negative. The current method does not claim jammer- or SIR-aware hard negative mining.

E. Objective and Curriculum

The full objective is

$$\mathcal{L} = L_{\text{mod}} + \lambda_j L_{\text{jam}} + \lambda_q L_q + \lambda_m L_{\text{mask}} + \lambda_x L_{\text{xcc}} + \lambda_\perp L_\perp, \tag{22}$$

where $L_{\text{mask}} = D_{\text{JS}}(M^\star \| M)$, L_q is validity-masked smooth L_1, and $L_\perp$ is the squared centered cosine similarity between active jammer-task and modulation-task embeddings. This orthogonality term is evaluated only for jammer-active views and does not establish source identifiability. The locked initial weights are $(\lambda_j, \lambda_q, \lambda_m, \lambda_x, \lambda_\perp) = (0.25, 0.05, 0.50, 0.10, 0.01)$ with label smoothing 0.05 and $\tau_x = 0.12$. Mask and contrastive terms ramp in after the classification warm-up. Learning-rate decay and checkpoint selection begin only after the complete objective has been active for a minimum number of epochs, preventing early validation checkpoints from bypassing the proposed losses.

V. Simulation Evidence Protocol

A. Data Layers

The evidence is intentionally layered.

1) A deterministic Python proxy generator is used for unit tests, diagnostics, and inexpensive screening. Its channel heuristics cannot support a standards-compliance claim.
2) The main simulation cache uses MATLAB `nrTDLChannel` primitives corresponding to TDL-A through TDL-E in 3GPP TR 38.901 [22]. Target and jammer waveforms traverse independent channel objects before shared receiver effects and exact realized SNR/SIR scaling. TDL-A/C/D are allocated to training and validation; TDL-B/E are reserved for a profile-held-out test.

TDL profiles alone do not instantiate every geometry, blockage, or nonstationary trajectory in a V2X evaluation methodology [23]. We therefore use the narrower term "TR 38.901 TDL-profile simulations." This paper makes no claim about SDR captures, field deployment, complete V2X scenario compliance, or engineering transfer.

B. Signal, Jammer, and Impairment Space

The locked target set contains BPSK, $\pi/2$-BPSK, QPSK, 8PSK, 16QAM, 64QAM, 256QAM, GMSK, CPFSK, and 4FSK. The seen jammer dictionary contains tone, multitone, chirp, sweep, partial-band, and comb waveforms; pulse and OFDM-like interference are held out by family. Modulated cochannel collisions and multi-jammer compositional mixtures are excluded from the single-label headline protocol. Here, *cochannel* means an in-taxonomy modulated cochannel collision, whereas *mixed* means a composition of multiple jammer families. Without a target anchor, swapping two in-taxonomy cochannel emitters can leave the observation distribution unchanged while changing the requested label. On a balanced source-exchange class with identical x and

TABLE I
LOCKED PROTOCOL SPLITS. "HELD" MEANS ABSENT FROM TRAINING.

Split	Jammer/composition	Speed	TDL profile
Train	seen + 20% clean	seen	A/C/D
Validation	seen + 20% clean	seen	A/C/D
ID test	seen + 20% clean	seen	A/C/D
Hard interference	seen, SIR$\leq$ 0	seen	A/C/D
Unseen jammer	held	seen	A/C/D
Unseen speed	seen	180/250 km/h	A/C/D
Held-out channel	seen	seen	B/E
Combined OOD	held	180/250 km/h	B/E
Clean retention	none	seen	A/C/D/B/E

two required labels, the mixture-only Bayes error is therefore at least $1/2$. Such collisions require a dominant-emitter label, an explicit physical anchor, or a separately scored ambiguity task. The exact discrete SNR grid for every split is $\{-10, -6, -2, 2, 6, 10, 14, 18\}$ dB. Jammer-active ordinary splits use SIR $\in$ $\{-10, -5, 0, 5, 10\}$ dB, while the hard-interference split uses SIR $\in$ $\{-15, -10, -5, 0\}$ dB. Clean samples have invalid SIR by construction. Training samples are drawn only from these declared discrete grids; there is no continuous training interpolation.

Receiver factors in the present implementation include carrier-frequency and phase offset, I/Q gain/phase imbalance, additive DC offset, and window-level AGC normalization. The multiplicative linear effects are shared across target, jammer, and noise so that the component identity in (2) remains auditable; DC is tracked in the unexplained component. Timing/sampling offset, receiver filtering, clipping, and quantization are outside the current evidence scope.

C. Factor-Isolated Splits

Table I prevents a combined shift from hiding its cause. Every split has independent source identities and generator/channel seeds. Training, validation, and ID testing use a deterministic 20% no-jammer quota; every stress split is jammer-active, whereas clean retention is entirely jammer-free. The combined set is a stress test, not a substitute for the three isolated tests.

D. Cache Audit

The offline builder pre-appends a fixed guard, batches MATLAB channel calls, audits returned filter and path delays, then crops the central window. It records target and jammer profile, delay spread, maximum Doppler, channel seed, guard margin, source ID, requested and realized SNR/SIR, component residual, file SHA-256, and a deterministic manifest digest. The eligible headline manifest is the sole source of reportable audit values; the current integration sentinels validate pipeline structure only and are not performance evidence.

VI. EXPERIMENTAL DESIGN

A. Comparators and Ablations

All trainable models receive identical source pairs, split manifests, epoch budget, optimizer family, label smoothing, and checkpoint policy. Literature models are labelled "reimplementation," "inspired," or "supervised adaptation" according to their provenance unless official code and the complete published protocol are executed unchanged. Quoted paper numbers are never mixed with unified retraining.

TABLE II
FAIL-CLOSED CACHE-AUDIT GATES BOUND TO THE ELIGIBLE HEADLINE MANIFEST; NO SENTINEL VALUE IS PROMOTED INTO THE MANUSCRIPT.

Gate	Required manifest evidence
Component identity	reload residual and tolerance
Power realization	SNR/SIR errors and validity masks
Guard adequacy	requested and post-delay margins
Immutability	file checksums and manifest digest
Source disjointness	pair/split source-ID intersections

Predeclared comparators include a classical higher-order/cyclostationary feature classifier, the direct shared backbone, MCLDNN [24], and a transparent IQFormer-inspired baseline. The recent low-SNR comparator is a CSSL-AMC supervised adaptation [14]: its encoder/classifier graph is audited against the authors' Apache-2.0 source at commit `2fbc5b3e12f7`, but it is randomly initialized and trained with the common paired-view classification objective, not the published two-stage protocol. It imports neither official weights nor reported results. MFENet [16] and RepCCNet [5] remain related work, not executed rows, because their reproducibility gate was not satisfied. Clean-input and oracle-teacher routes provide upper controls, not deployable baselines.

B. Metrics and Statistical Inference

Primary outcomes are macro-F1 and accuracy; secondary outcomes are worst supported-class recall, negative log-likelihood, and 15-bin expected calibration error. Results are reported over the full SNR–SIR plane and in the predeclared hard region, not only as a global mean.

The primary reference comparator is predeclared before any model training or test evaluation and is not selected on validation. For each frozen source set, models produce source-aligned probability bundles. Candidate–reference accuracy and macro-F1 differences use a 10,000-draw, class-stratified source-cluster paired bootstrap. For the five-seed headline comparison, the primary hierarchical paired bootstrap resamples both algorithm seeds and the same paired source clusters; source-only and seed-only intervals are also reported. An exact two-sided McNemar test is supplemental, restricted to one fitted-model seed, reported for every predeclared candidate, and never computed by pooling predictions across seeds. Holm adjustment is applied within each predeclared held-out-regime $\times$ seed family across the candidate model family; it does not correct across regimes or seeds and is not the basis for the headline inference. Validation is excluded from testing. Screening uses three model seeds, while the locked headline comparison uses five.

TABLE III
PRE-REGISTERED ABLATION CHAIN. "TEACHER" IS THE FIXED SIMULATION-COMPONENT POWER MASK TEACHER, AND XCC IS EXACT-SOURCE CROSS-CONDITION CONTRAST.

ID	Representation	Routes	Teacher	Jam/quality MTL	XCC	Bypass	Purpose
A0	Direct spectral backbone	–	–	–	–	–	shared-feature lower baseline; not parameter-matched
A1	Single sigmoid mask	1	–	–	–	–	ordinary soft selection
A2	Conditioned tri-mask	3	–	–	–	yes	third-route allocation and bypass without teacher
A3	Conditioned tri-mask	3	yes	–	–	yes	incremental physical supervision
A4	Conditioned tri-mask	3	yes	yes	–	yes	incremental branch semantics
A5	VIMD-Net	3	yes	yes	yes	yes	complete method
A6	Conditioned dual-mask	2	yes	yes	yes	yes	composite control; no route-count causal attribution
A7	Conditioned tri-mask	3	yes	yes	yes	–	bounded-bypass test

C. Mechanism Tests

Accuracy alone cannot verify the proposed interpretation. Let $W_m = M_s + \lambda M_o + \rho$. For independently tracked target and jammer spectra, define

$$\alpha = \frac{\|W_m \odot Z_s\|_2^2}{\|Z_s\|_2^2}, \qquad \beta = \frac{\|W_m \odot Z_j\|_2^2}{\|Z_j\|_2^2}, \tag{23}$$

$$\Delta_{\mathrm{spec}} = 10\log_{10}\frac{\alpha}{\beta}. \tag{24}$$

We report energy-weighted route correlation and MAE against the teacher, route occupancy, α, β, and Δ_{spec}. Because the additive bypass can make $W_m > 1$, α is a target-energy transfer ratio, not a bounded retention rate; its mean, maximum, and fraction of samples with $\alpha > 1$ are reported. The teacher's third route combines unexplained power and target–jammer power ambiguity, which are not identifiable as separate constituents from the student mask. Their teacher-side occupancies may be tabulated only to describe the simulated target; correlation and MAE are computed for the combined route. A 1,000-draw permutation test evaluates the diagnostic association between predicted and teacher third-route occupancy. These mechanism probes are exploratory diagnostic associations outside the headline multiplicity family. In particular, Δ_{spec} is an oracle-conditioned spectral component ratio, not waveform SIR gain, source separation, or a real-world SIR estimator.

The auxiliary heads are evaluated directly: jammer micro/macro F1 and AUROC, and validity-masked SNR, SIR, and Doppler MAE after conversion back to physical units. Undefined AUROC or unavailable scaling metadata is reported as unavailable rather than imputed.

D. Complexity

Parameter count and serialized state size are exact. Convolution/linear MACs exclude the STFT, whose real-operation count is reported separately from the standard FFT order estimate. Batch-one CPU and GPU P50/P95 latency are measured only in an isolated process after warm-up, with device, runtime, window length, and synchronization policy recorded. Latency obtained while another process occupies the GPU is rejected.

VII. RESULTS AND FALSIFICATION GATES

This section is intentionally unpopulated. The executable pipeline has passed unit, data-identity, cache, gradient-routing, and fixed-batch optimization checks, but the available smoke profiles are too small to estimate generalization. No ineligible diagnostic or screening run is promoted into Tables IV–VI.

TABLE IV
HEADLINE AMC RESULTS. ACC./F1/WCR ARE PERCENTAGES; LOWER NLL/ECE IS BETTER.

Model	Acc.	F1	WCR	NLL	ECE
Direct backbone			pending		
MCLDNN reimpl.			pending		
IQFormer-inspired			pending		
CSSL-AMC sup. adapt.			pending		
VIMD-Net			pending		

TABLE V
FACTOR-ISOLATED MACRO-F1 AND PAIRED GAIN OVER THE PRIMARY REFERENCE, PREDECLARED BEFORE TRAINING AND TEST EVALUATION.

Regime	Baseline	VIMD-Net	Gain [95% CI]
Hard SIR ≤ 0 dB		pending	
Unseen jammer		pending	
Unseen speed		pending	
TDL-B/E held out		pending	
Combined OOD		pending	
Clean retention A/C/D		pending	
Clean retention B/E		pending	

The method is rejected or narrowed if any of the following occurs: (i) the tri-mask does not outperform single- and dual-mask controls in the hard region; (ii) teacher supervision does not improve both mask agreement and downstream robustness; (iii) held-out gains disappear after paired multiplicity correction; (iv) clean-condition macro-F1 degradation exceeds 1.0 percentage points in point estimate or its paired 95% lower confidence bound falls below -2.0 percentage points. Clean retention is reported separately for seen-profile A/C/D and held-profile B/E strata; a predeclared aggregate may be descriptive but cannot replace either stratum. In these cases, failure is reported rather than repaired by deleting difficult classes or conditions.

VIII. REPRODUCIBILITY, LIMITATIONS, AND SCOPE

Every run stores the complete configuration, environment record, dataset manifest, digest, model seed, checkpoint, source-aligned probabilities, per-regime metrics, and paired statistics. Student initialization occurs after seeding. Validation selection is delayed until the full objective is active. Tables are

TABLE VI
MECHANISM AND COMPLEXITY OUTCOMES.

Quantity	VIMD-Net
Teacher–student mask JS $\downarrow$	pending
Third-route Spearman $\uparrow$	pending
Target-energy transfer ratio α	pending
Target amplification share $\Pr(\alpha > 1)$	pending
Jammer leakage $\beta \downarrow$	pending
Third-route diagnostic association	pending
Oracle spectral component ratio $\uparrow$	pending
Parameters	pending
Batch-one latency	pending

generated from eligible machine-readable artifacts; the paper has no manually typed performance cells.

Three limitations are structural. First, the teacher requires clean component bookkeeping during simulation training. It cannot be applied to unreferenced field captures, where mask supervision must be disabled. Second, TDL channels are standards-aligned primitives rather than a complete nonstationary V2X geometry; the manuscript therefore makes a simulation-only claim. Third, separately computed component STFT powers omit coherent cross terms and therefore define a simulation-component allocation target, not a conserved decomposition of mixture energy. The masks allocate task representations and do not reconstruct isolated time-domain sources; the additive bypass makes this distinction explicit.

IX. Conclusion

This paper formulates interference-robust AMC under vehicular-Doppler TDL-profile simulations as physically supervised representation allocation. VIMD-Net combines a target-power-dominant/jammer-power-dominant/ unexplained-or-power-ambiguous teacher, dual task branches, and exact-source cross-condition learning on a shared complex-STFT lattice. Its simulation claims are deliberately falsifiable through source-disjoint factor holds, single-view inference with source-paired statistics, and oracle-conditioned diagnostic component probes. The internal build establishes the method and locked protocol; quantitative conclusions will be stated only after the predeclared multi-seed evidence gate is satisfied. No SDR, field, deployment, or complete V2X-scenario conclusion is drawn.

Internal Submission Note

Before submission, the human authors must (i) replace authorship placeholders, (ii) verify every scientific claim and citation against the primary source, (iii) supply eligible locked simulation results without expanding the scope, (iv) set `\internalreviewfalse`, and (v) provide any disclosure required by the current IEEE/TVT policy for generative-AI-assisted language editing. An AI system cannot assume authorship or author responsibility.

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